3D PRINTING LAB

Course Code: 23A03607

B.Tech (ME) - III Year II Semester - JNTUA R23 Regulations

L-T-P-C: 0-1-2-3

LAB MANUAL

Name	
Roll No.	
Branch / Section	Mechanical Engineering
Academic Year	

Course Objectives

1. Understand different methods of 3D printing.
2. Gain knowledge about simulation of the FDM process.
3. Estimate time and material required for manufacturing a 3D component.

Course Outcomes

1. Explain different types of 3D printing techniques.
2. Identify parameters for powder binding and jetting process.
3. Determine effective use of ABS material for 3D printing.
4. Apply principles of mathematics to evaluate the volume of material required.

Syllabus Outline

Module	Content
1	Introduction to prototyping, working of a 3D printer, types of 3D printing machines. Exp 1: modelling of an engineering component and conversion to STL. Exp 2: slicing of STL file and study of effect of layer thickness, orientation and infill on build time. Exercise 1: Component-1; Exercise 2: Component-2.
2	Exp 1: 3D printing by varying layer thickness. Exp 2: by varying orientation. Exp 3: by varying infill.
3	Study of effect of different materials like ABS, PLA, resin etc. and dimensional accuracy.
4	Identifying the defects in 3D printed components.
5	Exp 1: modelling of a component using a 3D scanner for a real-life object of unknown dimensions (reverse engineering). Exp 2: 3D printing of the above modelled component.

Module 1: Introduction, Modelling and Slicing

Introduction to Prototyping

A **prototype** is a first working or visual model of a product used to check form, fit and function before mass production. **Rapid Prototyping (RP)** builds the part directly from CAD data, layer by layer, in hours instead of weeks. This process is now called **Additive Manufacturing (AM)** or 3D printing, as opposed to subtractive processes such as milling.

Working of a 3D Printer (FDM)

1. A 3D CAD model is exported as an STL file.
2. Slicing software cuts the model into thin horizontal layers and generates G-code (tool path, temperatures, speeds).
3. Filament is pushed by the extruder gears into the heated hot end (liquefier) and melted.
4. The molten material is deposited through the nozzle, tracing each layer on the heated bed, while the X-Y motion and the Z-axis step up by one layer height.
5. Each layer fuses with the layer below as it cools. Supports are printed for overhangs.
6. The part is removed and post-processed (support removal, sanding, vapour smoothing for ABS).

Types of 3D Printing Machines

Process	Principle	Material	Remarks
FDM / FFF	Melted thermoplastic extruded through nozzle	ABS, PLA, PETG, nylon	Cheap, common in labs
SLA	UV laser cures liquid photopolymer	Photo-resin	High accuracy, smooth surface
DLP	Projector cures a full layer at once	Photo-resin	Faster than SLA
SLS	Laser sinters powder bed	Nylon powder	No support needed
3DP / Binder jetting	Print head jets liquid binder onto a powder bed	Gypsum, sand, metal powder	Full colour possible; part is porous and needs infiltration
Material jetting (PolyJet)	Droplets of photopolymer jetted and UV cured	Photopolymers	Multi-material, fine detail
LOM	Sheets bonded and cut by laser/knife	Paper, plastic sheet	Low cost, large parts
SLM / DMLS	Laser fully melts metal powder	Steel, Ti, Al alloys	Dense metal parts

Powder binding and jetting parameters: binder saturation, layer thickness (typically 0.09-0.2 mm), drop volume and print-head speed, powder particle size, drying/curing time and infiltration.

Module 1, Exp 1: Modelling of an Engineering Component and Conversion to STL

Aim

To model an engineering component in a CAD package and convert it into an STL file for 3D printing.

Software

CATIA / SolidWorks / Creo / Fusion 360 (any solid modeller).

Theory

STL (Standard Tessellation Language / Stereolithography) represents the surface of a solid as a mesh of triangles. Each triangle stores three vertices and a unit normal pointing outward. It has ASCII and binary versions; it has no colour, units or units information. A finer export (smaller chord height / angular tolerance) gives a smoother surface but a larger file. A valid STL must be a **watertight (manifold)** mesh: no holes, no flipped normals, no self-intersections.

Procedure

1. Create a sketch and build the component using extrude, revolve, holes and fillets (e.g. a flanged bracket or gear blank).
2. Check dimensions and units (mm).
3. Use File > Save As / Export and choose STL; set the format (binary) and resolution (deviation / angle tolerance).
4. Open the STL in a viewer or the slicer and check for errors using the repair function.

Result

The component was modelled and converted into an STL file.

Viva Questions

1. Why is STL called a tessellated format?
2. What is a manifold mesh?
3. Effect of chord tolerance on STL quality?

Module 1, Exp 2: Slicing of STL and Effect of Process Parameters on Build Time

Aim

To slice an STL file and study the effect of layer thickness, orientation and infill on build time.

Software

Ultimaker Cura / PrusaSlicer / Simplify3D / Creality Slicer.

Theory

- **Layer thickness (h):** number of layers n = H / h. Halving h doubles the layers and roughly doubles the time, but improves surface finish (reduces the stair-step effect).
- **Orientation:** decides the part height (number of layers), amount of support, surface quality and strength (parts are weakest between layers, the Z direction).
- **Infill:** internal fill density (0-100 %) and pattern (grid, triangles, honeycomb, gyroid). Higher infill increases material, time and strength.
- **Other parameters:** nozzle diameter (0.4 mm), print speed, number of shells/walls, top/bottom layers, supports, adhesion (skirt/brim/raft).
- **Material estimate:** filament length $L = V / (\pi d^2/4)$ with d = 1.75 mm; mass = density x V. Typical density: ABS 1.04 g/cm^3, PLA 1.24 g/cm^3.
- **Time estimate (approx.):** time = total extrusion path length / print speed + travel and layer-change time.

Procedure

1. Import the STL into the slicer and check the size and position on the bed.
2. Set the material profile, nozzle, and initial values: layer 0.2 mm, infill 20 %, orientation flat.
3. Slice and note the estimated print time and filament length/weight.
4. Change only layer thickness (0.1, 0.2, 0.3 mm), re-slice and record.
5. Rotate the part to different orientations (flat, on side, upright), re-slice and record time, support volume and material.
6. Change infill (10, 20, 50, 100 %), re-slice and record.
7. Plot graphs of time against each parameter.

Observation Tables (fill from the slicer)

Layer thickness (mm)	Layers	Time (h:min)	Material (g)
0.10			
0.20			
0.30			

Orientation	Height (mm)	Support (g)	Time (h:min)
Flat			
On side			
Upright			

Infill (%)	Pattern	Time (h:min)	Material (g)
10	Grid		
20	Grid		
50	Grid		
100	Grid		

Result

Build time decreases with higher layer thickness and lower infill; orientation changes height and support, affecting both time and material.

Exercises

1. **Exercise 1 (Component-1):** model and slice the first assigned component; tabulate build time and material for the three parameters above.
2. **Exercise 2 (Component-2):** repeat for the second component and suggest the best parameter set (balance of time, strength and finish).

Module 2: 3D Printing by Varying Process Parameters

Aim

To 3D print a modelled component by varying (Exp 1) layer thickness, (Exp 2) orientation and (Exp 3) infill, and to compare the results.

Equipment

FDM 3D printer, PLA/ABS filament, slicer software, SD card/USB, digital vernier caliper, weighing balance, stopwatch.

General Procedure

1. Level the bed and clean it; load the filament; preheat nozzle and bed (PLA: 200 C nozzle, 60 C bed; ABS: 240 C nozzle, 100 C bed - as per the filament datasheet).
2. Slice the STL with the chosen parameter set and save the G-code.
3. Print and record actual print time; remove the part carefully.
4. Measure the weight and key dimensions; inspect surface finish.
5. Repeat for each parameter value.

Exp 1: Varying Layer Thickness

Layer (mm)	Actual time	Weight (g)	Surface finish (obs.)
0.1			
0.2			
0.3			

Expected: thinner layers give smoother surfaces and finer detail but take much longer.

Exp 2: Varying Orientation

Orientation	Actual time	Support used	Surface / strength (obs.)
Flat (0 deg)			
Inclined (45 deg)			
Vertical (90 deg)			

Expected: upright prints need more layers and have weaker inter-layer strength in the load direction; overhang faces need supports and show rougher finish where supports touch.

Exp 3: Varying Infill

Infill (%)	Actual time	Weight (g)	Remarks (rigidity)
10			
30			
60			
100			

Expected: time and weight increase with infill; strength rises, but with diminishing returns above about 50-60 %.

Result

The effect of layer thickness, orientation and infill on print time, weight and quality was studied.

Viva Questions

1. Which parameter affects surface finish most?
2. Why is a part weaker along the Z direction?
3. What is the role of infill pattern?

Module 3: Effect of Materials and Dimensional Accuracy

Aim

To study the effect of different materials (ABS, PLA, resin etc.) on the 3D printed part and its dimensional accuracy.

Material Comparison

Property	ABS	PLA	Photo-resin (SLA/DLP)
Process	FDM	FDM	Vat photopolymerisation
Nozzle temp. / bed	230-250 C / 90-110 C	190-220 C / 20-60 C	Cured by UV (about 405 nm)
Strength / toughness	Tough, impact resistant	Stiff but brittle	Brittle (some tough grades)
Shrinkage / warping	High; needs enclosure and heated bed	Low	Small, but cure shrinkage exists
Surface finish	Moderate; can be vapour smoothed with acetone	Good	Excellent, fine detail
Heat resistance	About 100 C	About 55-60 C	Varies
Fumes / safety	Styrene fumes; ventilate	Mild, bio-based	Resin is skin irritant; gloves, IPA wash
Typical use	Functional parts	Concept models, education	Detailed models, jewellery, dental

Effective use of ABS: use an enclosed chamber and heated bed (about 100 C), a brim or raft for adhesion, no cooling fan or very low fan speed, and good ventilation, to avoid warping and layer splitting.

Procedure

1. Print the same test cube (e.g. 20 x 20 x 20 mm) with ABS, PLA and resin using identical layer thickness.
2. Measure length, width and height at three places using a vernier caliper.
3. Compute the dimensional error and percentage error for each material.

Error (mm) = measured - nominal | % error = (measured - nominal) / nominal x 100

Material	Nominal (mm)	L (mm)	W (mm)	H (mm)	% error
ABS	20				
PLA	20				
Resin	20				

Result

The effect of material on strength, finish and dimensional accuracy was studied; resin gave the best accuracy and ABS showed the most shrinkage.

Module 4: Identifying Defects in 3D Printed Components

Aim

To identify defects in 3D printed components, find their causes and suggest remedies.

Defect	Appearance	Common causes	Remedy
Warping	Corners lift off the bed	Uneven cooling, poor adhesion, high shrinkage (ABS)	Heated bed, brim, enclosure, glue/PEI sheet
Stringing	Thin hair-like strands between features	High nozzle temp., wet filament, low retraction	Lower temp., increase retraction, dry the filament
Under-extrusion	Gaps, thin or missing walls	Clogged nozzle, low temp., wrong flow	Clean/replace nozzle, raise temp., calibrate flow
Over-extrusion	Blobs, rough surface, dimensions too large	Flow rate too high	Reduce flow, calibrate e-steps
Layer shifting	Layers displaced sideways	Loose belts, too high speed or acceleration	Tighten belts, reduce speed
Delamination	Layers split apart	Low temp., too much cooling, thick layers	Raise temp., reduce fan, use thinner layers
Elephant foot	Bulged first layer	Bed too hot, nozzle too close	Raise Z-offset, lower bed temp.
Poor overhangs / support scars	Drooping or rough surfaces	Insufficient support or cooling	Add supports, increase cooling, change orientation
Pillowing / gaps in top layer	Holes on top surface	Low infill, too few top layers	More top layers or infill
Stair-stepping	Visible steps on curved surface	Thick layers	Thinner layers, adaptive layers, sanding

Procedure

1. Visually inspect the printed samples from Modules 2 and 3.
2. Photograph or sketch the defects, classify them, and write the probable cause.
3. Propose and, if possible, test the corrective parameter change.

Result

The defects in the printed components were identified and remedies suggested.

Module 5: Reverse Engineering using 3D Scanner

Aim

(Exp 1) To model a real-life object of unknown dimensions using a 3D scanner (reverse engineering), and (Exp 2) to 3D print the modelled component.

Theory

Reverse engineering recovers a CAD model from a physical object. A **3D scanner** captures the surface as a **point cloud** - by laser triangulation, structured (white) light, or photogrammetry. Multiple scans are aligned and merged, converted to a polygon mesh (STL), cleaned and, if required, converted to a parametric CAD solid.

Exp 1: Procedure

1. Select an object; spray a matte developer if it is shiny or transparent; fix reference markers/targets on or around it.
2. Calibrate the scanner and scan the object from several angles (or on a turntable) with sufficient overlap.
3. Align (register) the scans and merge into a single point cloud.
4. Clean noise and outliers, fill holes, smooth and decimate the points.
5. Generate the mesh (triangulation) and check for errors.
6. Scale the model, measure key dimensions, and optionally build NURBS surfaces / solid features in CAD (e.g. Geomagic, MeshLab, Fusion 360).
7. Export as STL.

Exp 2: 3D Printing of the Modelled Component

1. Import the STL in the slicer; scale if needed and orient the part.
2. Select material and parameters (layer 0.2 mm, infill 20 %), add supports if required.
3. Slice, print and post-process.
4. Compare the printed part with the original object and calculate the dimensional deviation.

Result

A model of the object was obtained by 3D scanning and the component was 3D printed.

Viva Questions

1. What is a point cloud?
2. Why are reference markers used during scanning?
3. Name the steps from scan data to printed part.

References

1. Ian Gibson, David W. Rosen, Brent Stucker, *Additive Manufacturing Technologies: Rapid Prototyping to Direct Digital Manufacturing*, 1/e, Springer, 2010.
2. Chua C.K., Leong K.F. and Lim C.S., *Rapid Prototyping: Principles and Applications*, 2/e, World Scientific Publishers, 2003.